

Tensor Analysis

4. Contravariant, Covariant, and Mixed Tensors

Suppose N^2 quantities A^{qs} in a coordinate system $(x^1, x^2, \dots, x^N)$ are related to N^2 other quantities $\bar{A}^{pr}$ in another coordinate system $(\bar{x}^1, \bar{x}^2, \dots, \bar{x}^N)$ by the transformation equations

$$\bar{A}^{pr} = \sum_{s=1}^N \sum_{q=1}^N \frac{\partial \bar{x}^p}{\partial x^q} \frac{\partial \bar{x}^r}{\partial x^s} A^{qs} \quad p, r = 1, 2, \dots, N$$

Or

$$\bar{A}^{pr} = \frac{\partial x^p}{\partial \bar{x}^q} \frac{\partial x^r}{\partial \bar{x}^s} A^{\mu s}$$

by the adopted conventions, they are called contravariant components of a tensor of the second rank or rank two.

The N^2 quantities $A_{\bar{q}}$ are called covariant components of a tensor of the second rank if

$$\bar{A}_{ju} = \frac{it^0}{2\bar{x}^0} \frac{dx^3}{\partial x^r} A_\mu$$

Similarly, the N^2 quantities A_s^4 are called components of a mixed tensor of the second rank if

$$\bar{A}_r^p = \frac{\partial x^p}{\partial x^0} \frac{\partial x^s}{\partial x^r} A_s^q$$

Kronecker Delta

The Kronecker delta, denoted by δ_k^i , is defined as follows:

$$\delta_k^j = \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases}$$

As its notation indicates, it is a mixed tensor of the second rank.